COMPANION TO

Capability Matters: A Casebook

LENS Companion

CONCENTRATION · CROSSWALKS · REFERENCES

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Learning Engineering for Next-Generation Systems · v2.1

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LDT · SCHOOL OF EDUCATION · JOHNS HOPKINS UNIVERSITY

SECTION

How to use this companion

The LENS Companion is the concept-facing reference for the LENS specialisation: what LENS is, what it teaches, and how the casebook's cases line up with the curriculum. It travels with advisory boards, prospective students, recruiting conversations, and program reviews — anyone for whom the question is **what is this concentration?** rather than **where in the casebook is case 137?**.

The companion is in three parts:

I. The Five LENS Competencies — formal names, taglines, main objectives, and the v2.1 subobjective set (1.1 – 5.6). The organising layer over the formal CLOs.

II. The CLOs and the Course Mapping — the five concentration learning objectives, the LDT / LENS course descriptions, the course-to-CLO coverage matrix, and the LEN course list with per-course case counts.

III. Crosswalks — the casebook's induced framework mapped to the canonical five competencies, plus the three-anchor convention every v2 case carries.

> **For auditors and reviewers.** The casebook's indexes — every case by primary domain, every case by LENS course — and the full per-case references appendix live in a separate document: **Validation and Audit** (`validation-audit.typ` in the casebook source; `capability-matters-validation-audit.pdf` in the build). That document is built from the same casebook source as the companion, so the two stay in sync; it is the place for the verification and accreditation track.

> **On versioning.** This companion documents the **v2.1** program state (June 2026), in which D3 is renamed **Human-System Collaboration** and moved from position 5; T&E moves to D4; Sociotechnical Constraints moves to D5. The order now reads as the flywheel: see the system → build → integrate humans → measure → deploy. Prior versions are preserved in the program docs' change logs (`lens_program/1_LENS_Five_Competencies.md`).

SECTION

Why a companion — the thesis in brief

Learning engineering is a toolbox, not a single tool. At its centre is a dialogue between two traditions — the engineering disciplines that design, build, and sustain complex systems, and the learning sciences and education that study how people come to know and do. Around that dialogue gather many other fields: cognitive and human-factors engineering, measurement, implementation science, design, data and computation, and the operational domains themselves. The work is plural by design.

Capability is a system parameter.

Every complex system depends on what people can do inside it. That dependency is measurable, designable, and too important to leave to chance.

> **01.** Systems don't fail because the technology breaks. They fail because someone couldn't do what the system required.
>
> **02.** We engineer every parameter of a system except the people.
>
> **03.** Capability is not a soft problem. It is a systems engineering problem.
>
> **04.** Decision-grade evidence. Not opinions about training.
>
> **05.** Capability without agency is automation. The goal is operators who can perform, adapt, and lead — not comply.
>
> **06.** Sometimes the constraints make the goal unreachable. Saying so early is a result, not a failure.

Capability lives at the **interface** between what a system requires of its operators and the impact the system has to deliver. When that interface is wrong, the gap may originate in human development, in system design, in organisational performance, or in the interaction among them. Distinguishing which is itself a measurement and engineering problem; the casebook calls it **gap attribution**, and treats it as the diagnostic move that the rest of the discipline depends on.

The premise of the Johns Hopkins School of Education's Learning Design & Technology program — and of the Learning Engineering for Next-Generation Systems specialisation that has grown out of it — is that the capability parameter is engineerable. Not by accident. Not by declaration. By the same kind of evidence-grounded, methods-grounded, cross-domain discipline that built modern reliability engineering, modern epidemiology, and modern systems safety. The discipline does not yet exist at the scale the evidence demands. LDT and LENS are organised to help build it.

> **The casebook's method, in one sentence.** Each case is an **unpacking**: take a real outcome, recover the capability question behind it, and attribute the gap to human development, system design, or organisational performance. The discipline of unpacking is what LENS teaches; the casebook is its primary instrument. The framework that follows is the analytic lens the unpacking method uses — five competencies, each a way of asking what the work the system requires actually is, and whether the people inside it can do it.

SECTION

The Five LENS Competencies — v2.1

Organising layer over the formal CLOs. Formal names carry program documentation; taglines carry recruiting and the site. Subobjective numbers (1.1, 3.2, …) are stable tags for case studies, course modules, and capstone rubrics. The CLO language in the program documentation remains the language of record.

1 Systems Analysis

SYSTEMS THINKING AND ANALYSIS (CLO-1)

see the whole system

Main objective. Trace how human capability and system performance depend on each other, so interventions target the real problem.

1.1 Decompose system performance requirements into measurable human capability requirements: define what the system demands of its operators and the impact it must deliver.

1.2 Model learning environments and their host operational systems as interacting systems, with defined components, interfaces, and feedback loops.

1.3 Apply systems engineering lifecycle models to scope, sequence, and evaluate capability interventions within and across disparate systems.

1.4 Analyze and communicate human-system interdependencies to identify capability gaps and predict operational impact at scale.

1.5 **Governance-objection diagnostic.** Distinguish a governance objection that good design can dissolve from one that correctly signals the system should not deploy. [v2]

2 Iterative Development

LEARNING ENGINEERING DESIGN AND IMPLEMENTATION (CLO-2)

build, test, refine

Main objective. Design capability interventions through iterative engineering cycles that survive contact with real operational environments.

2.1 Design interventions that integrate learning sciences principles with measurable outcomes and system design constraints.

2.2 Run the iteration cycle: design, instrument, evaluate, refine, and revise the problem framing as evidence accumulates.

2.3 Evaluate evidence-based design strategies for transfer to high-consequence settings at operational speed and scale.

2.4 Construct implementation plans that address adoption, sustainment, and lifecycle integration across organizational and system boundaries.

2.5 **Narrate and defend the design iteration in first person.** Design competence includes not only producing an outcome but rendering the iteration legible. [v2]

3 Human-System Collaboration

HUMAN-SYSTEM COLLABORATION AND ADAPTIVE SYSTEMS (CLO-3)

work together

Main objective. Design human-system partnerships — including but not limited to human-AI — that make people more capable while preserving human agency and the recoverability of the system.

3.1 Design role architectures, interface and alert systems, mode/state transparency, authority gradients, and recoverability mechanisms that engineer collaborative capability at the human-system boundary.

3.2 Evaluate the measured impact of system-mediated work on human performance: the gains the system enables and the risks it introduces — automation bias, cognitive offloading, skill atrophy.

3.3 **Delegation with revocation.** Design the human oversight layer for delegated work; specify in advance the disconfirming evidence that would revoke the delegation. [v2]

3.4 **Collaboration measurement.** Measure the capability of a team or collaboration as a unit of analysis, distinct from any individual operator or any individual system component. [v2]

3.5 Specify learning and capability requirements for systems not yet fielded, working from design artifacts rather than operational experience.

4 Test and Evaluation

DATA, MEASUREMENT, AND EVALUATION (CLO-4)

show what works

Main objective. Produce decision-grade evidence linking learning to operational impact — where decision-grade is a sufficiency judgment under irreducible uncertainty — and tell a learning gap from a system gap.

4.1 Design ethical instrumentation strategies that measure capability at the speed and scale operations require.

4.2 **Gap attribution.** Diagnose the gap: differentiate capability shortfalls rooted in human development from those rooted in system design or organizational performance.

4.3 Construct decision-grade evidence artifacts under irreducible uncertainty: state what is known, what is assumed, and what would change the decision.

4.4 **Judgment under inadequate evidence.** Justify a consequential decision on incomplete and contested evidence; document the basis; name what would change it. [v2]

4.5 Communicate evidence honestly to technical and non-technical stakeholders, including uncertainty and limits of inference.

4.6 **Fairness beyond omission.** Demonstrate that omitting a protected attribute does not establish fairness; analyze competing fairness definitions using demographic-stratified outcome evidence. [v2]

5 Navigating Sociotechnical Constraints

CONTEXT AND DOMAIN FLUENCY (CLO-5)

make it work in the real world

Main objective. Integrate capability interventions into the sociotechnical systems they must live in: the regulatory, organizational, cultural, and technical realities that determine whether good designs survive.

5.1 Analyze the regulatory, organizational, cultural, and technical constraints that shape what can be built, deployed, and sustained in a given context.

5.2 Apply human systems integration frameworks to fit capability interventions to operational environments.

5.3 Leverage prior domain expertise, and respect for others', to elicit and validate specialist knowledge that surfaces constraints early.

5.4 Translate constraints and stakeholder requirements across disciplinary and institutional boundaries into coherent design specifications.

5.5 Anticipate adoption and sustainment barriers, and design interventions that survive them.

5.6 **Cross-regime / platform-dependency governance.** Where capability is deployed on a platform governed by a different regime than the one operating it, design the governance seam as an explicit deliverable. [v2]

See the whole system. Build, test, refine. Work together. Show what works. Make it work in the real world.

SECTION

CLOs and Course Mapping — v2.1

The LDT M.Ed. is a 10-course, 30-credit, online/asynchronous master's degree in the School of Education. LDT comprises three concentrations: Learning Experience Design (LXD, currently operating), AI Leadership in Education (AILE, launching August 2026), and Learning Engineering (LENS, launching August 2026). LENS graduates demonstrate competency across five domains; the CLOs below are the assessable form.

THE FIVE CLOS

CLO-1 · Systems Thinking and Analysis. Decompose system performance requirements into measurable human capability requirements. Apply systems engineering lifecycle models. Analyze interdependencies between human and system performance to identify gaps and predict operational impact at scale.

CLO-2 · Learning Engineering Design and Implementation. Design capability development interventions through iterative engineering cycles. Evaluate evidence-based design strategies for transfer to high-consequence settings. Construct and communicate implementation plans.

CLO-3 · Human-System Collaboration and Adaptive Systems. Design human-system partnerships that make people more capable while preserving human agency. Specify delegation with revocation. Measure collaboration as a unit of analysis.

CLO-4 · Data, Measurement, and Evaluation. Design ethical instrumentation. Construct decision-grade evidence under irreducible uncertainty. Apply gap attribution. Demonstrate fairness beyond omission.

CLO-5 · Context and Domain Fluency. Integrate capability interventions into regulatory, organizational, cultural, and technical realities. Synthesize across boundaries. Anticipate adoption and sustainment barriers, including cross-regime / platform-dependency governance.

COURSE-TO-CLO COVERAGE

Course	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
LDT F1 (Learning Sciences Studio)		✘			
LDT F2 (Critical Perspectives)					✘
LEN 1 (Principles of LE for Complex Systems)	✘	✘			
LEN 2 (Human-AI Teaming and Adaptive Learning)			✘		
LEN 3 (Learning Engineering Systems)	✘				✘
LEN 4 (Evidence, Analytics, Measurement)				✘	
LEN 6 (Applied Problem Solving in LE)	✘	✘			✘
LEN 10 (Capstone)	✘	✘	✘	✘	✘
LEN 5 (Human Capability Analysis, elec.)	○				○
LEN 7 (Bias, Risk, Governance, elec.)					○

LEN 8 (Knowledge Transfer & Org Learning, elec.)		○			○
LEN 9 (Computational and AI Methods, elec.)			○	○	

✗ = required course; ○ = elective. Foundations + concentration + methods + capstone = 8 required courses (24 credits); 2 electives complete the 30-credit degree.

SECTION

Crosswalks

INDUCED 8 (CASEBOOK SCAFFOLD) → CANONICAL 5 (PROGRAM OF RECORD)

The casebook induces an 8-cluster analytic scaffold from its 194 cases (full account: `competencies.md`). Those eight cluster cleanly into the canonical five LENS competencies — except for D2 Iterative Development, which is the iteration **method** threaded through the cases rather than its own cluster. The gap is what the editor-commissioned first-person Practice Flywheel accounts will close in the next edition.

Induced cluster (casebook)	Canonical D	Notes
1 Capability requirements under operational reality	D1	Engineered vs. stated requirements; sustainment-as-requirement.
2 Evidence architecture the institution cannot deceive itself with	D4	Measurement gaming; post-deployment surveillance.
3 Interface and role design at the human-automation boundary	D3	Cues, alerts, mode/state transparency, recoverability.
4 Pairing mechanism with authorization	D3	Frontline halt authority; non-punitive reporting; authority gradient.
5 Governance architecture for deployment	D5	Consent, oversight, accountability as design artifacts.
6 Cross-organization and cross-time knowledge transfer	D5	Institution-building after catastrophe; tacit-knowledge survival.
7 Capability under system change and aging assumptions	D5	Legacy assets; multi-layer drift; transitions.
8 Equity and construct definition as design commitments	D4	Construct choice; demographic stratification; fairness.
(none specific)	D2	The iteration **method** — threaded across cases, not its own cluster.

THE CASEBOOK'S THREE ANCHORS PER CASE

Every v2 case carries three anchor codes in its header and metadata:

`induced-anchor` — e.g. `"2.4"` — the casebook's induced 8 / 32 sub-cluster the case evidences (analytic scaffold).

`lens-anchor` — e.g. `"D3/PT5"` — the canonical LENS five domain + problem type (1 – 6) the case is **primary** for.

`clo-anchor` — e.g. `"CLO-3"` — the assessable CLO the case serves as a worked example for (course mapping anchor).

APPENDIX · CLOS AND COURSE COVERAGE

The competencies, the CLOs, and the courses.

COMPETENCY → CONCENTRATION LEARNING OUTCOME

The five capabilities the cases turn on are named, in the curriculum, as five concentration learning outcomes (CLOs):

Systems Analysis	CLO-1 · Systems Thinking and Analysis
Iterative Development	CLO-2 · Learning Engineering Design and Implementation
Test and Evaluation	CLO-3 · Data, Measurement, and Evaluation
Navigating Sociotechnical Constraints	CLO-4 · Context and Domain Fluency
Human-System Collaboration	CLO-3 · Human-System Collaboration and Adaptive Systems

THE LEN COURSES · CASES PER COURSE

Each case names the LEN courses it serves as a worked example for, and most cases serve several. The counts below are therefore not a partition of the 202 cases — they sum to well more than 202, because a single case typically appears under three or four courses.

LEN 1	Principles of learning engineering	36
LEN 2	Human-AI teaming	54
LEN 3	Systems integration	49
LEN 4	Evidence and measurement	67
LEN 5	Capability analysis	86

LEN 6	Applied problem-solving	18
LEN 7	Bias and governance	114
LEN 8	Knowledge transfer	98
LEN 9	Computational methods	30
LEN 10	Capstone studio	19

WHAT THE DISTRIBUTION SHOWS

The distribution is itself diagnostic. The strongly covered topics are the ones the existing case literature most readily supports: bias and governance (LEN 7) and knowledge transfer (LEN 8) lead, with capability analysis (LEN 5) and evidence and measurement (LEN 4) close behind. Across 202 cases the corpus is saturated with examples of what goes wrong when ethics, governance, evidence, and institutional learning are not engineered — and, in the success cases, what it takes to engineer them.

The thinner coverage is equally informative. Computational methods (LEN 9) has the failure modes of algorithmic systems well represented but the computational tools for capability instrumentation under-represented. The capstone (LEN 10) maps to fewer cases than the number of studio projects each cohort undertakes. Systems integration (LEN 3) and applied problem-solving (LEN 6) required focused tagging to reach defensible coverage, surfacing that operational systems-engineering content is under-represented in the published case literature relative to its curricular centrality, and that stakeholder-analysis and problem-framing have not historically been packaged as case material by other disciplines. These are pedagogical gaps the LENS studio sequence is organized to close — not by adding more failure cases, but by having students produce the missing artifacts in real operational settings and entering them into the dataset for the next cohort.

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